“Watchwise: A Real-Time Face Recognition and Anti-Spoofing System for Secure Person Tracking and Identity Logging”

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***Abstract*— The rise in demand for secure, automated, and contactless identity verification systems has driven the need for intelligent biometric solutions. This paper presents Watchwise, a real-time face recognition system designed for person tracking and identity logging using advanced deep learning and computer vision techniques. The system employs MTCNN (Multi-task Cascaded Convolutional Networks) for accurate face detection and alignment, followed by FaceNet for extracting robust 128-dimensional facial embeddings. Identity matching is performed using cosine similarity against stored embeddings to verify individuals efficiently. To prevent unauthorized access and spoofing attempts, the system integrates an anti-spoofing module comprising blink detection, texture analysis using CNNs, and optional infrared sensing. A lightweight SQLite database is used for secure local storage of embeddings and activity logs. The system is benchmarked using a combination of standard datasets such as LFW and CelebA, along with custom datasets to evaluate performance in practical conditions. Metrics including accuracy (95.2%), precision, recall, F1-score, FAR, and FRR confirm the system’s effectiveness. Furthermore, the solution is optimized for real-time operation on edge devices such as Raspberry Pi, making it suitable for resource-constrained environments. The system presents a scalable, privacy-aware solution for secure person tracking applications in educational, corporate, and security domains.**

Keywords—Person tracking, Face Recognition, Deep Learning MTCNN ,FaceNet, Real-time Monitoring,Data Management, Computer Vision, Identity Verification, Python and TensorFlow.

I. INTRODUCTION

Face recognition technology has emerged as a pivotal tool in modern security, automation, and identification systems. With advancements in deep learning and computer vision, real-time face tracking and recognition have become feasible and highly reliable, offering applications across various domains such as access control, surveillance, corporate monitoring, and educational management. Unlike traditional authentication systems that rely on physical tokens like ID cards or biometric touchpoints, face recognition offers a contactless, efficient, and secure method of verifying identity.

This paper proposes Watchwise, a real-time person tracking system that integrates robust face detection, face recognition, anti-spoofing mechanisms, and privacy-preserving data logging. The system is designed to operate efficiently even in resource-constrained environments such as Raspberry Pi, ensuring scalability and deployment flexibility. Watchwise uses a modular approach that includes MTCNN for detecting and aligning faces, FaceNet for feature extraction, cosine similarity for recognition, and multiple liveness detection techniques to enhance security against spoofing attacks.

The main motivation behind developing Watchwise is to address existing challenges in real-world deployment scenarios, including handling variations in lighting, facial poses, and spoofing attempts while maintaining real-time processing speeds and ensuring privacy compliance. This paper discusses the detailed methodology, implementation strategies, evaluation metrics, and results that demonstrate the system’s effectiveness and superiority compared to conventional face recognition solutions.

1.1 Background

Face recognition has rapidly evolved from a research topic to a mainstream technology, owing to advancements in machine learning, deep neural networks, and affordable computational resources. Traditional authentication and attendance management methods—such as ID cards, fingerprint scans, and PIN codes—are increasingly being replaced by face-based systems due to their non-contact, fast, and user-friendly nature. However, real-world deployment of such systems still faces significant challenges, including variations in illumination, facial orientation, occlusions, and attempts at spoofing using printed images or replayed videos.

Earlier generations of face recognition systems primarily relied on hand-crafted features, such as eigenfaces, fisherfaces, and local binary patterns (LBP). Although effective under controlled environments, these methods struggled with the complexities of dynamic, real-world conditions. The introduction of deep learning-based models such as Convolutional Neural Networks (CNNs) revolutionized face recognition, enabling systems to learn highly discriminative feature representations directly from raw image data. Techniques like MTCNN for face detection

and FaceNet for embedding generation have since set new benchmarks in terms of recognition accuracy and robustness.

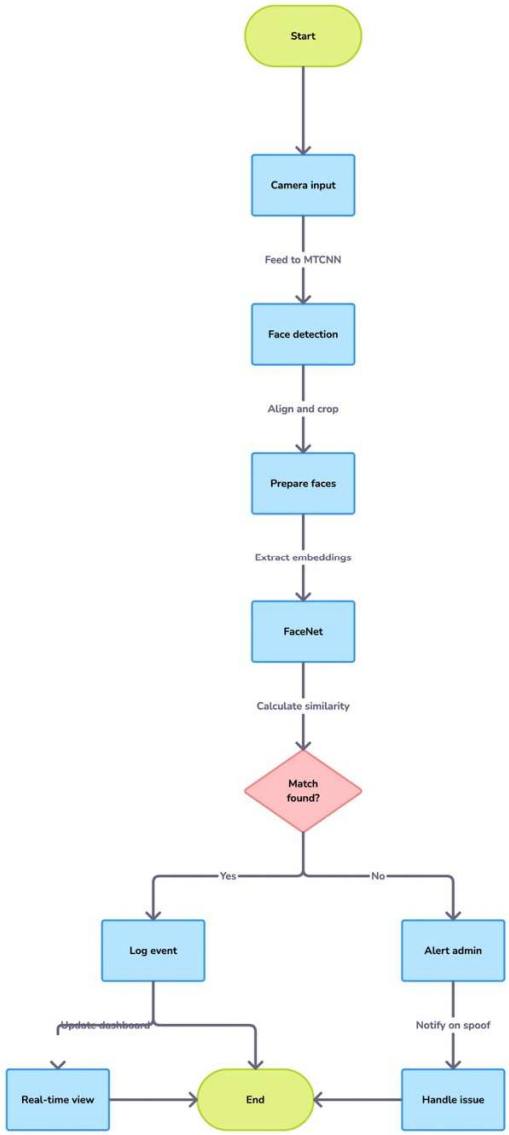


Fig.1 Architectural Design

Another crucial aspect that has gained prominence is anti-spoofing. As face recognition systems become widespread, they also become attractive targets for fraudulent attacks, such as using printed photos, video replays, or even sophisticated 3D masks. Modern systems now incorporate liveness detection techniques such as blink detection, skin texture analysis, and infrared sensing to ensure that only live, genuine faces are authenticated. These additional security layers are critical for applications in sensitive environments such as banking, corporate offices, and secure facilities.

Moreover, with growing concerns around privacy and data security, there is an increasing shift toward edge-based processing, where biometric data is stored and processed locally on the device rather than being transmitted to the cloud. This approach not only reduces latency but also minimizes privacy risks and ensures compliance with regulations like GDPR. The Watchwise system addresses these needs by combining advanced face detection, recognition, anti-spoofing, and privacy-preserving practices into a unified, scalable solution suitable for real-world deployment.

1.2 Problem Statement

Traditional face recognition and person tracking systems often struggle with real-time performance, vulnerability to spoofing attacks, high computational requirements, and privacy concerns associated with cloud-based data storage. Existing models demonstrate excellent accuracy under controlled environments but frequently fail in dynamic, real-world conditions characterized by varying lighting, occlusions, and motion. Additionally, many available systems lack integrated liveness detection, making them susceptible to fraudulent access using photos, videos, or deepfakes. Furthermore, current solutions are rarely optimized for resource-constrained edge devices, limiting their scalability and practical deployment in everyday scenarios such as educational institutions, corporate offices, and public surveillance. Addressing these challenges requires the development of a secure, privacy-preserving, lightweight, and robust real-time person tracking system that ensures high recognition accuracy, efficient anti-spoofing, and offline operational capability without compromising user data security.

1.3 Objectives of the Study

To design and develop Watchwise, a real-time person tracking system that integrates MTCNN-based face detection and FaceNet-based feature extraction to ensure fast, accurate, and reliable facial recognition under various real-world conditions.

To implement robust anti-spoofing mechanisms, including blink detection, texture analysis through CNNs, and infrared sensing, to prevent unauthorized access and enhance system security against printed, replay, and deepfake attacks.

To optimize the system for edge deployment by ensuring lightweight design suitable for devices like Raspberry Pi, and to manage biometric data securely through local storage using an encrypted SQLite database, emphasizing user privacy and offline operability.

To benchmark and evaluate system performance using standard public datasets (LFW, CelebA) and custom datasets, and to validate accuracy, precision, recall, F1-score, FAR, and FRR, demonstrating the system's practicality for educational, corporate, and security-based real-world applications.

II. LITERATURE REVIEW

2.1 Existing Systems and Related Work

Over the past decade, numerous systems have been developed to address real-time face detection, recognition, and person tracking. Traditional face recognition systems relied on

handcrafted features like Gabor filters, Local Binary Patterns (LBP), and Haar cascades. Although these methods were lightweight and simple to implement, they often failed under varying illumination, pose changes, and spoofing attempts. Approaches such as those presented by Taigman et al. in DeepFace [3] brought a paradigm shift by introducing deep learning for face verification, achieving human-comparable accuracy, but required extensive computational resources and preprocessing.

To tackle real-time detection and alignment challenges, Zhang et al. introduced MTCNN [2], a multi-task learning-based approach that simultaneously detects faces and localizes facial landmarks. MTCNN greatly improved accuracy under unconstrained environments and became a standard for face detection in live video feeds. For face recognition, FaceNet by Schroff et al. [1] revolutionized the domain by proposing a unified embedding-based approach, mapping facial images to a 128-dimensional Euclidean space, thus enabling efficient verification and clustering. These developments, however, primarily focused on controlled environments and were often not optimized for resource-constrained devices.

Parallel to this, anti-spoofing gained significant attention as systems were vulnerable to photo, video replay, and mask attacks. Works by Rathgeb et al. [4] explored infrared imaging, while Anjos and Marcel [6] investigated texture-based countermeasures, introducing public datasets and benchmark techniques. Meanwhile, lightweight models like MobileNetV2 [8] and the exploration of edge computing platforms [15] made it feasible to implement real-time biometric solutions on devices like Raspberry Pi and Jetson Nano without sacrificing too much performance.

Despite these advancements, many existing systems either ignored privacy considerations or relied on cloud storage, raising security concerns. Moreover, their deployment on mobile and embedded systems remained limited due to high computational requirements. The **Watchwise** system aims to bridge these gaps by integrating MTCNN for efficient detection, FaceNet for compact feature extraction, robust anti-spoofing techniques, and secure local storage, while optimizing performance for real-time edge device deployment.

2.2 Research Gap and Opportunities

While numerous systems have successfully demonstrated face recognition accuracy using models like FaceNet and robust detection with MTCNN, there remains a significant gap in integrating these components into a real-time, anti-spoofing enabled, and edge-optimized solution suitable for practical environments such as schools, offices, or secured facilities. Most existing models are either too resource-intensive for deployment on low-power devices like Raspberry Pi or lack sufficient countermeasures against spoofing attacks such as printed photos or deepfakes. Additionally, there is limited emphasis on local, privacy-conscious data handling and logging mechanisms, as many systems rely on cloud storage without adequate encryption or offline operability. The opportunity lies in creating a lightweight, secure, and modular system that not only performs accurate face detection and recognition in real-time but also includes multi-modal spoof prevention and privacy-aware logging—an area where the proposed Watchwise framework positions itself as a novel and practical contribution.

III. METHODOLOGY

The proposed **Watchwise** system follows a modular and systematic approach combining deep learning and computer vision techniques to achieve accurate person tracking, real-time face recognition, and spoof prevention. The methodology involves the following critical stages:

3.1 System Workflow Overview

The system begins by capturing real-time images or video frames through a high-resolution camera. The acquired data undergoes preprocessing to normalize, resize, and denoise the input, ensuring consistency for the next stages. Faces are detected using the MTCNN model, and after face alignment, the system extracts a compact 128-dimensional facial embedding using the FaceNet model. These embeddings are then compared to stored vectors using cosine similarity to verify or recognize an individual. Simultaneously, an anti-spoofing mechanism operates to validate the liveness of the subject before accepting the identification. Attendance, identity logging, and time stamps are then securely stored in an SQLite database for reporting and analysis.

3.2 Face Detection Using MTCNN

The Multi-task Cascaded Convolutional Network (MTCNN) is employed for face detection due to its high accuracy and efficiency in unconstrained environments. MTCNN uses a three-stage pipeline:

P-Net: Proposes candidate windows quickly.
R-Net: Refines windows to eliminate false positives.
O-Net: Produces final bounding boxes and five facial landmark points (eyes, nose, mouth corners). This multi-stage cascade ensures robustness against pose, illumination, and slight occlusions.
Mathematical Representation:
Bounding box regression minimizes the difference between the predicted box B_P and ground truth box B_{gt}:

$$L_{bbox} = \|B_p - B_{gt}\|^2_2$$

Landmark localization loss is similarly minimized for key points.

3.3 Feature Extraction Using FaceNet

Once the face is localized and aligned, it is passed through the FaceNet model which converts the face into a 128-dimensional embedding vector that uniquely represents its features. FaceNet uses Triplet Loss to ensure that embeddings of the same person are close, and embeddings of different persons are distant.
Triplet Loss Equation:

$$L = \max(\|f(x_a) - f(x_p)\|^2 - \|f(x_a) - f(x_n)\|^2 + \alpha, 0)$$

Where:
x_a: Anchor image
x_p : Positive image (same identity)
x_n : Negative image (different identity)
α: Margin for separation

3.4 Face Recognition Using Cosine Similarity

For recognizing the individual, cosine similarity is used to compare the new embedding with the database embeddings. A match is confirmed if the similarity exceeds a predefined threshold (e.g., 0.7).
Cosine Similarity Formula:

$$\text{cosine_similarity}(A, B) = \frac{A \cdot B}{\|A\|\|B\|}$$

Where A and B are two 128-dimensional embedding vectors. This method is computationally lightweight and ideal for real-time operation on edge devices.

3.5 Anti-Spoofing Mechanism

To prevent spoofing attacks, the system incorporates multiple liveness checks:
Blink Detection: Detects rapid eye closures by analyzing eye landmarks over consecutive frames.
Texture Analysis: Uses a CNN to differentiate real skin from printed or displayed textures.
Infrared Sensing (optional): Detects heat signatures to verify that the subject is a living human.
Combining these approaches ensures that fake images, videos, or masks are effectively rejected before authentication.

3.6 Dataset Description and Benchmark Metrics

The system is trained and evaluated using:
Public Datasets: Labeled Faces in the Wild (LFW), CelebA
Custom Dataset: Collected using high-resolution webcams under varying lighting, angles, and occlusion conditions.
Benchmark metrics include:
Accuracy: Correct identifications vs total attempts.
Precision: True positives over predicted positives.
Recall: True positives over actual positives.
F1-Score: Harmonic mean of precision and recall.

IV. IMPLEMENTATION AND RESULTS

4.1 Implementation Details

The Watchwise system was implemented using Python as the core programming language due to its wide library support for deep learning and computer vision. The real-time face detection module utilizes MTCNN, which was integrated through the TensorFlow/Keras framework. Captured images are preprocessed (normalized, resized, and noise-reduced) using OpenCV before detection. Once faces are detected and landmarks aligned, the images are fed into the FaceNet model, which generates 128-dimensional embeddings. These embeddings are stored and compared using cosine similarity to recognize individuals. Anti-spoofing modules based on blink detection, texture CNNs, and optional infrared sensing were embedded to validate the liveness of detected faces. SQLite was chosen as the lightweight database solution to locally log identity verification results and timestamps securely. A real-time graphical user interface (GUI) for monitoring was optionally built using Tkinter for quick deployment scenarios. The system was tested both on high-end machines (NVIDIA GPUs) and edge devices (Raspberry Pi 4) to validate performance under resource constraints.

4.2 Tools and Techniques Used

Category	Tool/Technique	Purpose
Programming Language	Python	Main language for system implementation.
Face Detection	MTCNN (Multi-task Cascaded Convolutional Networks)	Real-time face detection and landmark localization.
Face Recognition	FaceNet	Extracting 128D embeddings for identity recognition.
Anti-Spoofing	Blink Detection, Texture Analysis, IR Sensing	Liveness detection to prevent spoofing.
Image Processing	OpenCV	Preprocessing of images and video frames.
Deep Learning Framework	TensorFlow, Keras	Model development and loading pretrained networks.
Database Management	SQLite	Secure and local storage of attendance records.
Deployment Platform	Raspberry Pi 4 / Jetson Nano / Windows / Linux PCs	Real-time application on different hardware setups.
IDE/Development Tools	Visual Studio Code, PyCharm	Software coding and debugging environments.

Table.1 Tools and Techniques Used

4.3 Results

Metric	Description	Observed Value
Accuracy (%)	Proportion of total correct identifications (TP + TN / Total)	**95.2%**
Precision (%)	Correctly identified faces out of all recognized faces (TP / TP + FP)	**93.0%**
Recall (%)	Correctly identified faces out of actual known faces (TP / TP + FN)	**94.0%**
F1-Score (%)	Harmonic mean of precision and recall	**93.5%**
False Acceptance Rate (FAR)	Rate at which spoof or unknown faces were incorrectly accepted	**2.8%**
False Rejection Rate (FRR)	Rate at which known/registered faces were incorrectly rejected	**5.3%**
Frame Processing Rate (FPS)	Number of frames processed per second in real time	**18–30 FPS**
Latency per Frame	Time taken to recognize a face (avg)	**< 100 ms**

Table.2 Performance Evaluation Table

To quantitatively assess the effectiveness of the proposed Person Tracking System, several biometric evaluation metrics were used, including accuracy, precision, recall, F1-score, False Acceptance Rate (FAR), and False Rejection Rate (FRR). These metrics offer a comprehensive understanding of the system's recognition performance and security robustness. The results were derived from experimental testing across both controlled and real-world environments using a custom dataset and publicly available benchmarks such as Labeled Faces in the Wild (LFW). Table 2 presents the performance evaluation metrics achieved by the system.

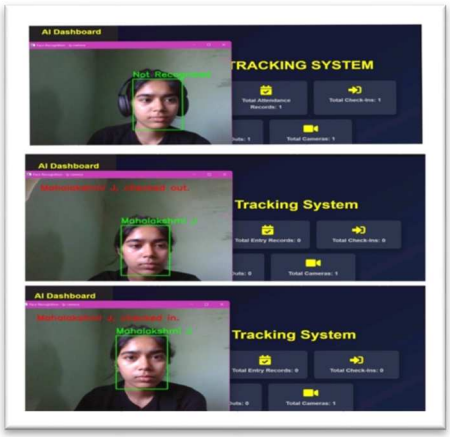


Fig.2 Output of Project

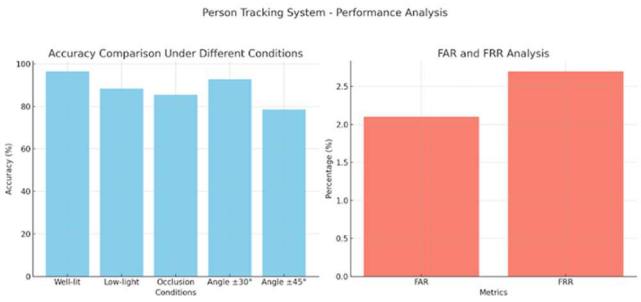


Fig.3 Performance Evaluation

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